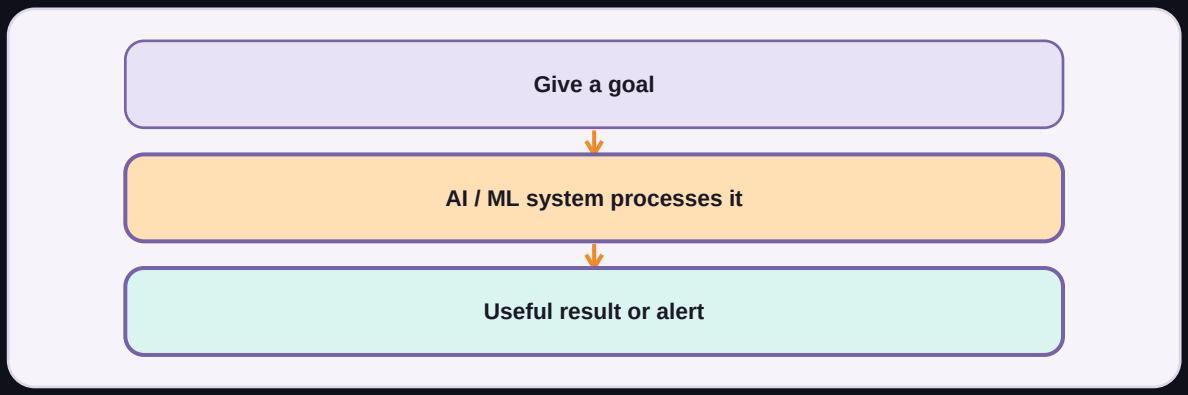
HACK-O-WEEN

AI Topics Explained Visually

Nine topics, simple diagrams, and a few practical uses for each.



How to read this guide
Each topic has one simple diagram, a few points explaining what it means, where it is used, and one example to help you imagine it. The first two topics also include Halloween-themed hackathon ideas.

Inspired by the dark haunted-house look of your Hack-o-Ween template.

1. Phantom Agents

Hackathon track: Agentic AI

What it means: An AI system that can plan and complete a sequence of tasks using tools, rather than only replying with text.

VISUALISE THE WORKFLOW

What it does

- Plans steps to reach a goal.
- Uses tools such as APIs, files, search, or approved code.
- Checks results and can recover from some errors.

Where it is used

- Research and report creation
- Coding assistants and business workflows
- Cybersecurity and personal AI assistants

Picture it: You ask for a report. One agent finds information, another writes, and a verifier checks it.

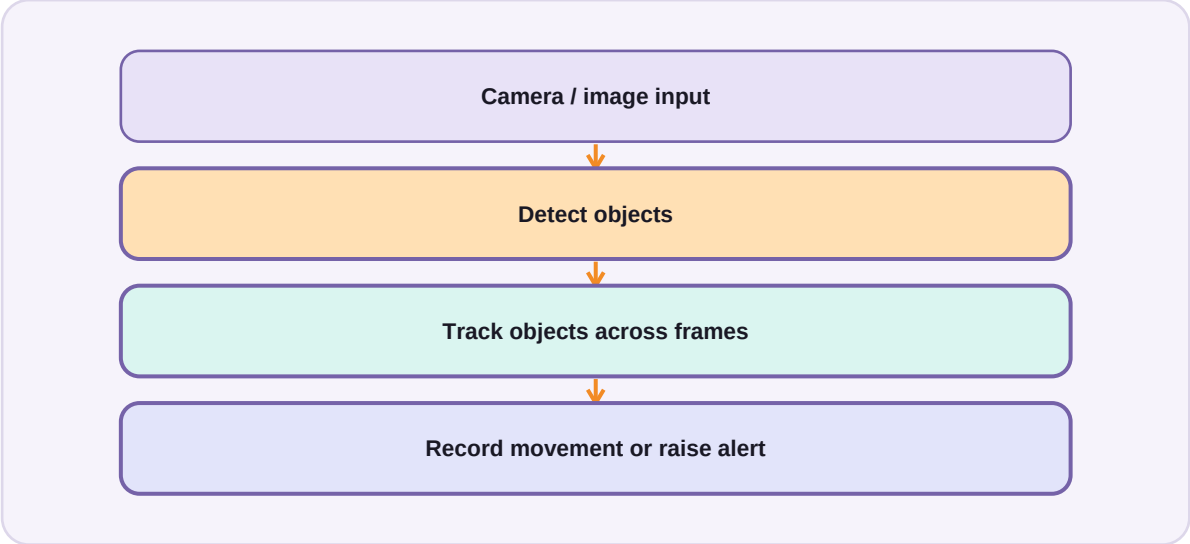
Halloween idea — The Haunted Protocol: Agents explore a digital haunted mansion. A security agent detects a malicious 'cursed' instruction and blocks it while the mission continues safely.

2. Voodoo Vision

Hackathon track: Computer Vision & Spatial Computing

What it means: Technology that helps computers interpret images and videos, locate objects, and follow how they move.

VISUALISE THE WORKFLOW



What it does

- Object detection identifies what an object is and where it appears.
- Object tracking associates the same object across video frames.
- Spatial computing reasons about position and relationships in space.

Where it is used

- Security and surveillance
- Robotics and autonomous vehicles
- Medical image analysis, AR/VR, and industrial monitoring

Picture it: A camera detects a bottle, draws a box around it, then follows it as someone moves it.

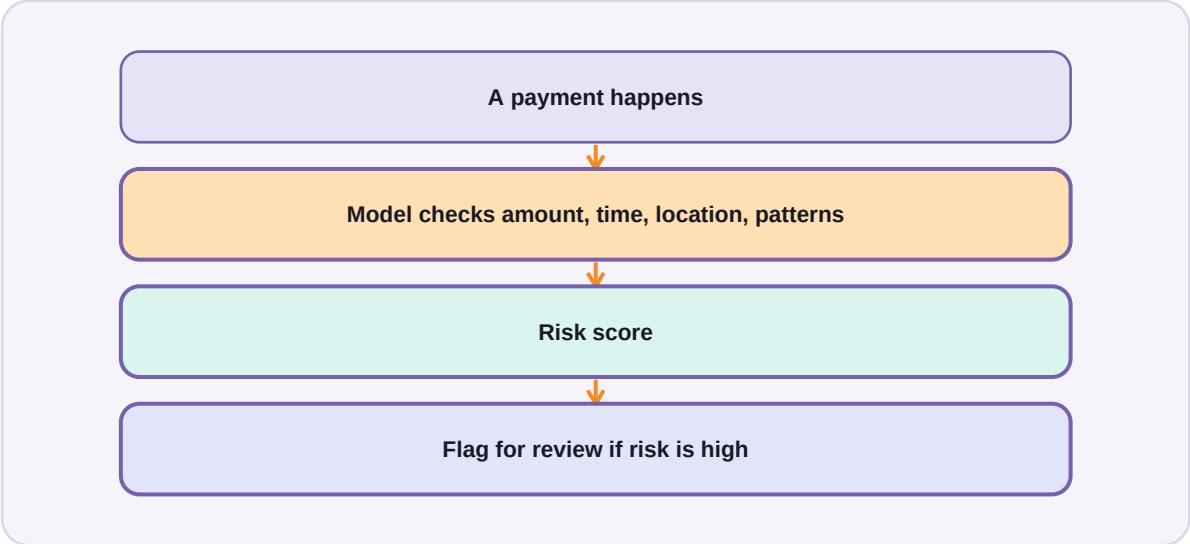
Halloween idea — The Phantom Lens: Track a skeleton prop, ghost decoration, or pumpkin around a haunted-house scene. Log when it moves or disappears.

3. The Vampire's Ledger

Hackathon track: FinTech

What it means: AI and data tools that help detect suspicious payments and make financial activity safer.

VISUALISE THE WORKFLOW



What it does

- Learns patterns from suitable past transactions.
- Highlights unusual transactions for review.
- Can display a risk score and reasons for an alert.

Where it is used

- Bank and payment-app fraud detection
- Credit-card security
- Loan risk and stock-market analysis

Picture it: Several unusually large payments appear in a short time, so the system flags them for review.

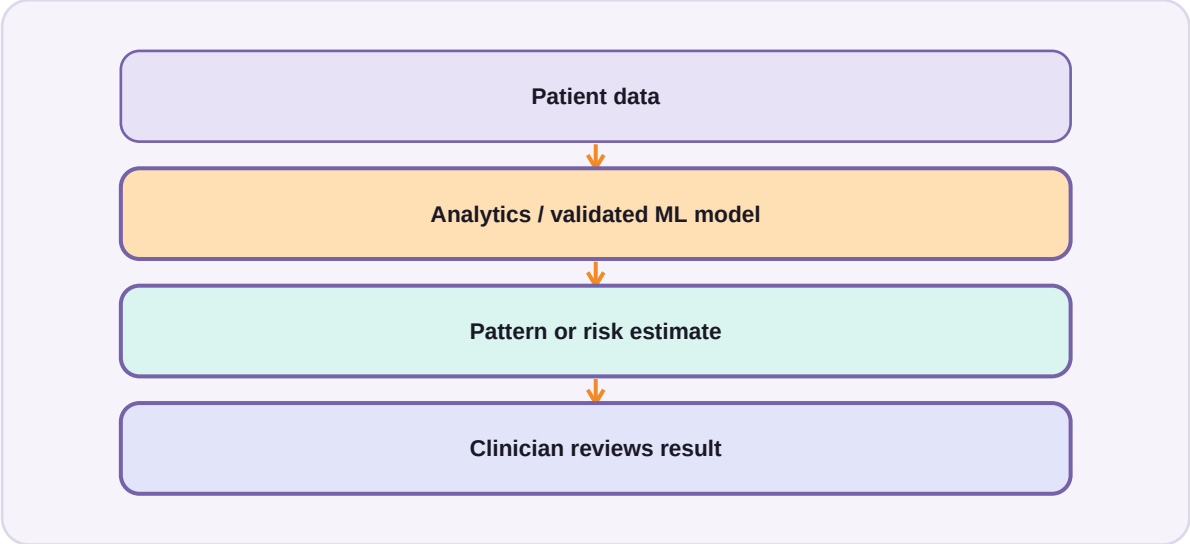
A flagged transaction is not automatically fraud. Human review and good data are important.

4. The Alchemist's Lab

Hackathon track: HealthTech

What it means: AI and software that help health professionals analyse medical information and spot patterns that may need attention.

VISUALISE THE WORKFLOW



What it does

- Analyses measurements, test results, or medical images.
- Highlights patterns and estimates risk when appropriately validated.
- Supports clinicians rather than replacing medical judgement.

Where it is used

- Medical-image analysis
- Patient monitoring and early-warning dashboards
- Clinical research and hospital operations

Picture it: A dashboard notices a concerning change in simulated patient measurements and asks a clinician to review it.

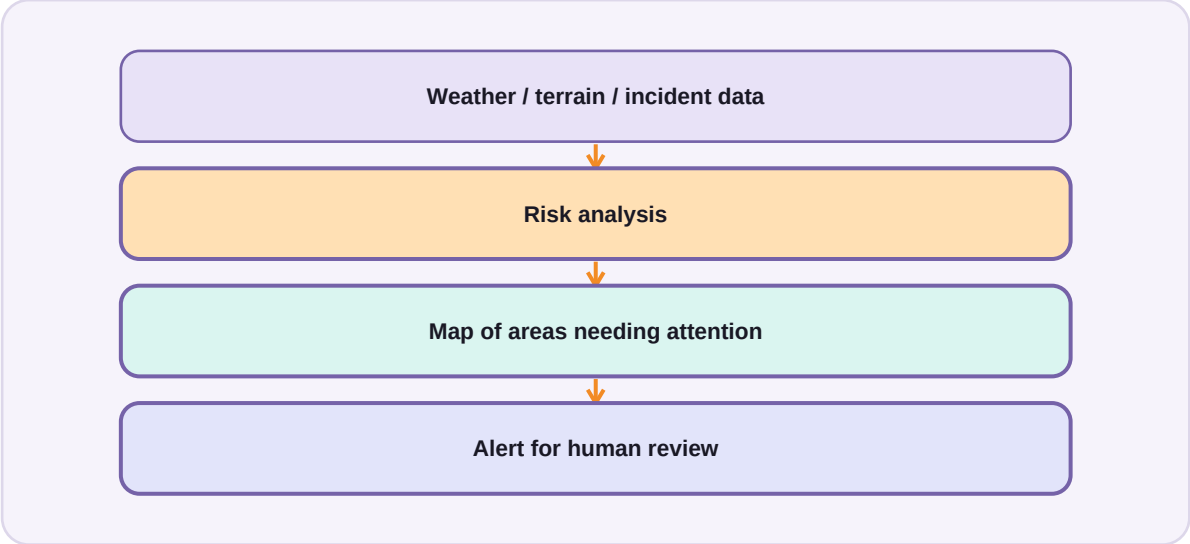
Medical AI needs appropriate data, privacy protections, validation, and careful wording. A prototype is not a proven diagnosis.

5. The Last Lantern

Hackathon track: Climate & Disaster Tech

What it means: Data-driven systems that analyse environmental hazards and support earlier warnings and emergency response.

VISUALISE THE WORKFLOW



What it does

- Combines information such as rainfall, terrain, and past incidents.
- Estimates which places may face higher risk.
- Shows findings on a map or alert dashboard.

Where it is used

- Flood and wildfire risk analysis
- Weather and environmental monitoring
- Emergency planning and smart-city tools

Picture it: A map highlights locations that may be more flood-prone based on rainfall and historical records.

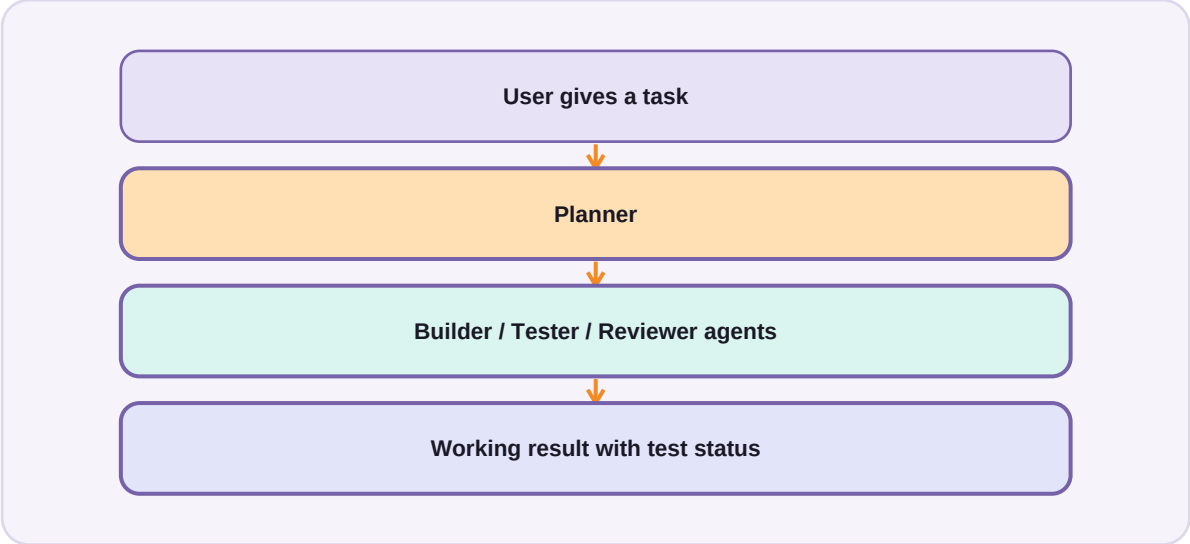
Risk estimates need reliable data and testing; a prototype should not claim to predict disasters accurately without evidence.

6. Phantom OS

Personalised project idea: An autonomous AI workspace

What it means: A product where several agents work together to complete a user's goal, such as building and testing a small application.

VISUALISE THE WORKFLOW



What it does

- Turns one goal into smaller tasks.
- Lets specialised agents build, test, and review.
- Reports what worked and what failed.

Where it is used

- Software-development helpers
- Research automation
- Personal productivity and internal business tools

Picture it: Ask it to create a document-analysis app. It plans the steps, builds a prototype, runs tests, and shows test results.

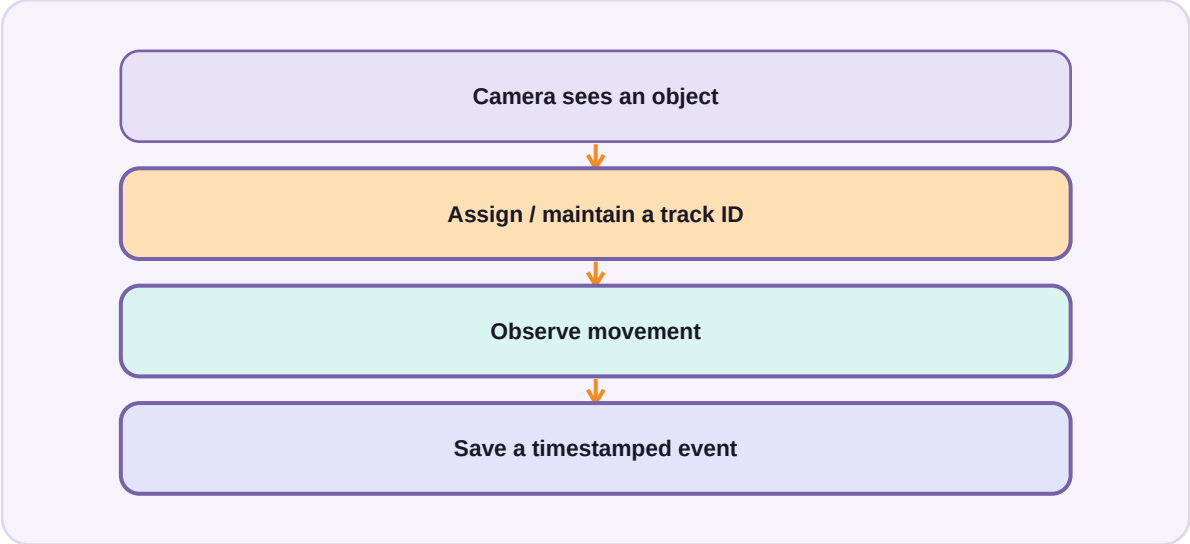
Keep code execution sandboxed and permissioned. Start with one workflow that can be verified.

7. RealityTrace

Personalised project idea: Visual memory and object tracking

What it means: A computer-vision system that tracks objects over time and keeps a timeline of what it observed.

VISUALISE THE WORKFLOW



What it does

- Detects objects and follows them across frames.
- Stores observations and timestamps.
- Builds a searchable record of observed movement.

Where it is used

- Inventory and smart-room monitoring
- Robotics and visual inspection
- Behaviour or event analysis in recorded video

Picture it: A phone is seen on a desk, picked up, then placed elsewhere. The system records the observed movement.

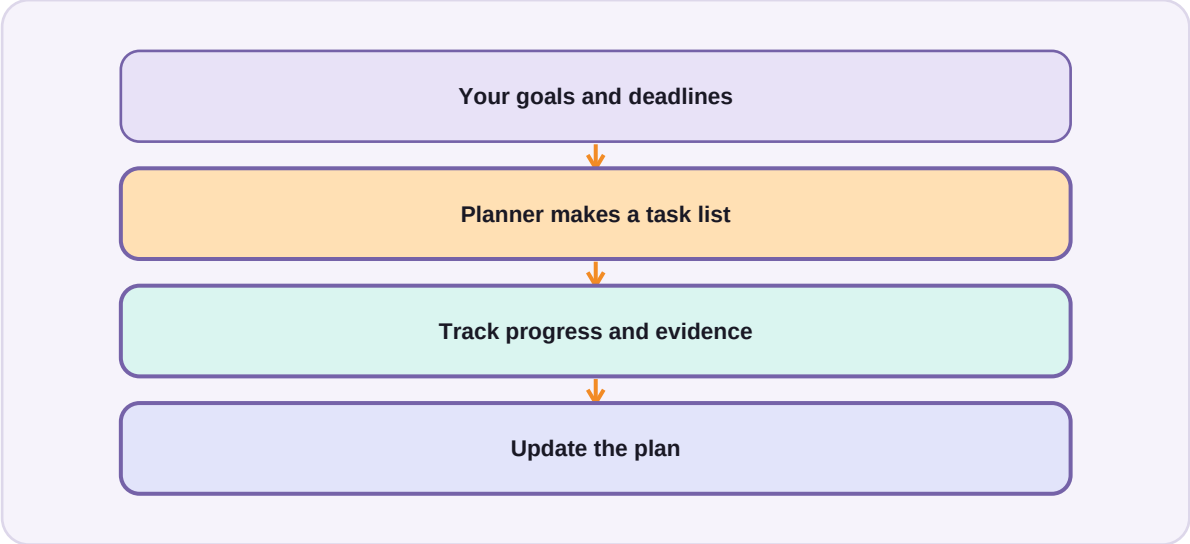
A camera cannot see behind an opaque door or inside a drawer. The system should only report what its views or other sensors support.

8. LifePilot AI

Personalised project idea: A task and goal assistant

What it means: An assistant that helps plan work, track progress, and adjust a plan when deadlines or task status change.

VISUALISE THE WORKFLOW



What it does

- Breaks large tasks into smaller steps.
- Shows what is done, delayed, or next.
- Can adapt a plan when real task information changes.

Where it is used

- Student study planning
- Project and team work management
- Personal goal tracking

Picture it: You enter exams and projects. It creates tasks, tracks progress, and reshuffles priorities when a task runs late.

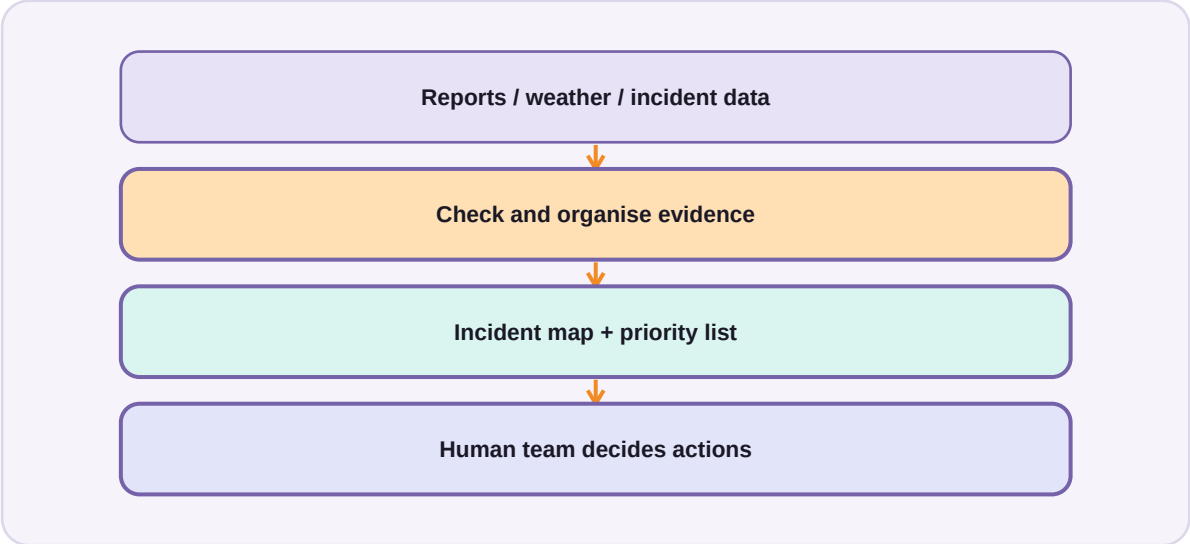
It becomes more useful when connected to real task data or calendars, instead of only generating plans.

9. Sentinel AI

Personalised project idea: Disaster-response assistant

What it means: A system that organises incident information and helps people prioritise response tasks.

VISUALISE THE WORKFLOW



What it does

- Brings different incident reports together.
- Helps sort incidents by stated priority and evidence.
- Shows locations and response tasks in one place.

Where it is used

- Emergency coordination
- Disaster-management dashboards
- City operations and incident-response teams

Picture it: Several flood reports arrive. The system groups them by location and produces a map and task list for responders.

Sentinel focuses on coordinating responses; The Last Lantern focuses more on environmental risk analysis and prediction.